

Deterrence and Deescalation: U.S. Strategies for a Nuclear-Capable Iran

Working Draft

University of Chicago Existential Risk Laboratory

Simon Siskel

August 16, 2024

1 Introduction

The prospect of Iran achieving a nuclear weapons capability poses a profound challenge to the security and stability of the Middle East. Iran's nuclear ambitions have long been a source of international concern, prompting a range of diplomatic efforts and economic sanctions aimed at preventing the development of nuclear weapons. However, should these efforts fail, the region and the world would face a drastically altered security landscape.

A nuclear-capable Iran might not only embolden Tehran's regional ambitions but would fundamentally shift the deterrence calculus for the United States and its allies. The potential for nuclear escalation, direct conflict, and increased proxy warfare necessitates a thorough examination of possible scenarios and strategic responses. This paper seeks to provide a comprehensive analysis of these dynamics, with a focus on understanding the implications for U.S. policy and regional stability.

Central to this analysis is the intricate network of Iranian proxies, including groups like Hezbollah, Iraqi Shia militias, and the Houthis in Yemen. These groups extend Iran's influence across the region, complicating efforts to contain Tehran's ambitions and posing direct threats to U.S. interests and allies. The relationship between the United States and Israel is also a critical component of this discussion. The strategic partnership between these two nations, while robust, is characterized by a lack of a formal mutual defense pact, raising questions about the extent of U.S. commitments in times of crisis.

This paper will explore various scenarios that the U.S. might face in dealing with a nuclear-capable Iran, from peacetime and crisis situations to limited conventional conflicts and the potential breach of the nuclear taboo. By examining these scenarios and their implications, the paper aims to provide actionable policy recommendations to enhance U.S. and Israeli security, reinforce deterrence, and maintain regional stability.

In light of the complex and volatile nature of Middle Eastern geopolitics, it is imperative for the United States to adopt a multifaceted and flexible approach. This includes bolstering defense commitments, enhancing military preparedness, and ensuring that diplomatic efforts are underpinned by credible and robust deterrence strategies. The recommendations provided herein seek to guide U.S. policy in navigating the challenges posed by a nuclear-capable Iran, ensuring that the security interests of the United States and its allies are safeguarded in an increasingly uncertain regional landscape.

This paper makes several key assumptions which frame the analysis and policy recommendations presented:

1. *Iran's Nuclear Ambitions*

Iran is actively pursuing a nuclear weapons capability, despite international efforts to curtail

its nuclear program through diplomatic and economic measures. Tehran has ambitions of not just remaining in a state of nuclear latency, but also of eventually achieving a nuclear weapons capability.

2. Regional Instability

The Middle East will continue to be a region of significant geopolitical instability, with multiple state and non-state actors engaged in ongoing conflicts and power struggles.

3. U.S.-Israel Strategic Partnership

The United States will continue to maintain a strong and strategic partnership with Israel, characterized by military aid, joint defense initiatives, and diplomatic support. The United States will remain committed to the defense of Israel, and will remain interested in ensuring the survival of Israel.

4. Deterrence Dynamics

The acquisition of nuclear weapons by Iran would fundamentally alter the deterrence dynamics in the Middle East, affecting the security calculations of regional and global actors.

5. Proxy Warfare

Iran will continue to leverage its network of proxies and allied groups across the region to project power and influence, even with a nuclear deterrent.

6. U.S. Commitment

The United States is committed to maintaining regional stability and ensuring the security of its allies, particularly Israel, through a combination of military presence, diplomatic efforts, and economic sanctions.

2 How might the US deter Iran from achieving deployable nuclear weapons?

2.1 Background

More than 20 years ago, U.S. intelligence uncovered clandestine uranium enrichment facilities in Iran in violation of the Nuclear Nonproliferation Treaty (NPT). Since then, the U.S. and allies have devoted significant effort (through diplomatic, political, and military means) to prevent Tehran from gaining access to a nuclear weapons capability. Despite massive investment by the Iranian regime in moving ever-closer to such a capability, Tehran has not crossed the threshold of a nuclear weapons capability. However, Iranian leadership's continued investment in its nuclear

program, and policy of nuclear hedging signals its commitment to sustaining its nuclear ambitions, despite the corresponding economic and political costs of such actions. The 2015 Joint Comprehensive Plan of Action (JCPOA), albeit flawed, was the most significant agreement in limiting Iran's progress towards a nuclear weapons capability since it began its nuclear program. The 2018 decision of the Trump administration to withdraw from the JCPOA, and subsequent sputtering attempts at revitalization of the agreement by the U.S. under President Biden, have had important implications. Iran's leadership has ramped up its production of enriched uranium, installed more advanced and efficient centrifuges, and has denied many of the monitoring devices and officials of the International Atomic Energy Agency (IAEA) access to its nuclear program, all in violation of its commitments under the JCPOA. The U.S. and allies have limited policy options in deterring Tehran from pursuing, and eventually achieving, a nuclear weapons capability.

The U.S. decision to withdraw from the JCPOA further reinforced an existing tenet of the Iranian regime, that the West is no longer a preferable or viable partner economically. Iranian leaders were privy to the decline of Washington's influence and capabilities internationally, and have prudently facilitated a pivot towards partnership with Asian powers. Iran has built increasingly close relationships – both political and economic – with Moscow and Beijing. Tehran has entered major economic agreements with the People's Republic of China, including a \$400 Billion deal in July of 2021. Beijing has been willing to deny rising U.S. sanctions on Iran since 2018 after the United State's exit from the JCPOA, which has allowed for the survival of Iran's economy. Iran has also been able to regain access to its pre-sanctions oil exports through increased exports to China, which have reached their highest levels ever, and seem to be only increasing.

Tehran's relationship with Moscow has also grown stronger, especially after Russia's invasion of Ukraine in February 2022. Russia's relationship with the West has deteriorated throughout the conflict, allowing Iran to facilitate its bilateral relationship with Russia through a series of high-level government official visits, and increased economic, political, military, and energy cooperation. The partnership between Tehran and Moscow involves investment by Moscow in Iran's gas development and trade. Perhaps more significantly, ties between the two regimes have evolved into what President Biden characterizes as a "full-fledged defense partnership." The defense partnership offers significant military benefits to both states. Russia has received, and deployed, thousands of Iranian Unmanned Aerial Vehicles (UAVs) to target critical Ukrainian infrastructure. Iranian military is also helping train Russian soldiers in the use, development, and production of these and more advanced systems. Tehran may also supply Moscow with ballistic missiles to aid its war effort against Ukraine. In return, Moscow has promised to send fighter jets, helicopters, and new air defense systems to Iran. The export of such advanced systems from Moscow to Tehran has already begun.

Iran's strengthening ties with Moscow and Beijing present significant challenges to U.S. efforts

to deter Iran from achieving a nuclear weapons capability. These relationships can undermine U.S. sanctions, offer Iran alternative economic and military support, and limit diplomatic leverage. The primary tool the United States has used to deter Iran from developing a nuclear weapons capability has been a combination of economic sanctions and diplomatic pressure. The U.S. has imposed a variety of unilateral sanctions, including financial sanctions, trade sanctions, and secondary sanctions. Financial sanctions have largely targeted Iran's financial institutions, restricting their access to the global financial system, and freezing assets. Trade sanctions, as the name suggests, impose restrictions on trade, particularly on key sectors such as oil and gas, which are critical to Iran's economy. Finally some sanctions have targeted non-U.S. entities that do business with Iran, thereby extending the reach of U.S. sanctions globally. Each of the above unilateral sanctions will lose their potency as Iran continues to shift its economy towards the East, with strong economic support from Beijing and Moscow.

The U.S. and its allies have also imposed some multilateral sanctions against Iran, through the United Nations, the European Union, and other Allies. By working through the United Nations to impose sanctions, the U.S. adds international legitimacy and broadens the impact of such sanctions. Similarly, coordinating with allies, especially the European Union, to impose complementary sanctions enhances their effectiveness. Politically, strengthening alliances with regional partners such as Israel, Saudi Arabia, and the United Arab Emirates to present a united front against Iran's nuclear ambitions can be an effective approach. The U.S. has also engaged with platforms such as the United Nations and the IAEA to apply pressure on Iran. However, Russia and China have mitigated the effects of U.S. economic sanctions by increasing trade and investment with Iran. By buying Iranian oil, providing financial services, and supplying goods and technology that the U.S. has restricted, Russia and China have enabled Iran to persist despite a massive sanctions regime imposed by the U.S. Furthermore, close ties with Russia and China have helped Iran break out of diplomatic isolation, reducing the effectiveness of U.S. efforts to build an international coalition against Iran's nuclear program.

2.2 Iran's Nuclear Program: Status

What is the status of Iran's nuclear program today? What are the implications of the current status of Iran's nuclear program, and potential nuclear capabilities on a U.S. effort to deter Tehran from weapons capability? According to the Annual Threat Assessment of the U.S. Intelligence Community in 2024, "Iran is not currently undertaking the key nuclear weapons-development activities necessary to produce a testable nuclear device. Since 2020, however, Tehran has stated that it is no longer constrained by any JCPOA limits, and Iran has greatly expanded its nuclear program, reduced IAEA monitoring, and undertaken activities that better position it to produce a nuclear device, if it chooses to do so." A nuclear weapons capability requires fulfillment of three

major technological capabilities. These include enrichment of uranium to weapons-grade levels, "nuclear weaponization," and delivery.¹

Iran currently possesses a large stockpile of enriched uranium to 60 percent, which is just short of the 90 percent enrichment levels desired for making nuclear weapons (weapons-grade uranium). Iran could quickly make the jump from 60 percent enriched uranium to weapons-grade Uranium, and could possess enough fuel for at least a handful of nuclear bombs within months. Some estimates declare that Iran could have enough weapons-grade uranium for one nuclear bomb within a week, six within a month, and within 5 months, roughly twelve nuclear bombs. According to the Washington Post, "Iran already possesses a stockpile of about 300 pounds of highly enriched uranium that could be further refined into weapons-grade fuel for nuclear bombs within weeks, or perhaps days, U.S. intelligence officials say. Iran also is believed to have accumulated most of the technical know-how for a simple nuclear device, although it would probably take another two years to build a nuclear warhead that could be fitted onto a missile, according to intelligence officials and weapons experts."² Iran's ability to enrich uranium has improved in recent months, as Tehran ramps up initiatives to build more and advanced new centrifuges. Without dispute, Iran has the knowledge and technical prowess required to achieve a nuclear weapons capability. Iran's plans for expansion, according to State Department spokesman Matthew Miller, are enhancing its nuclear program "in ways that have no credible peaceful purpose."³ According to the Washington Post, while the IAEA was aware of Tehran's plans for expansion, the size of the boost took many analysts by surprise. The expansion plan includes two of Iran's most critical nuclear program sites, Fordow and Natanz. At Fordow, the expansion would double the number of working centrifuges at the facility within about a month. At Natanz, a smaller but still significant increase is also planned. According to David Albright, as cited in The Post, "the 1,400 advanced machines would increase Fordow's capacity by 360 percent," while "Natanz's overall production capacity would increase by 35 percent."⁴ The increase in capabilities at Fordow and Natanz is not only due to the installation of more operational centrifuges, but also a leap in the technology of the centrifuges installed. At Fordow, "only newer-model machines, known as IR-6s, were to be installed, reports show, a substantial upgrade from the IR-1 centrifuges currently in use there." At Natanz, the plans call for the installation of "thousands of centrifuge machines of a different type, known as the IR-2M."⁵ According to a conservative analysis, Fordow alone could generate about 320 pounds of weapons-grade uranium within a month, and as much as 500 pounds within two months. If fully

¹"Annual Threat Assessment of the U.S. Intelligence Community." 2024.

²Warrick, Joby. 2024. "Iran Signals a Major Boost in Nuclear Program at Key Site." Washington Post, June 19, 2024.

³Warrick, Joby. 2024. "Iran Signals a Major Boost in Nuclear Program at Key Site." Washington Post, June 19, 2024.

⁴Albright, David. "How Quickly Could Iran Make Nuclear Weapons Today?" Institute for Science and International Security.

⁵Albright, David. "How Quickly Could Iran Make Nuclear Weapons Today?" Institute for Science and International Security.

implemented, Iran would achieve an unprecedented capability to produce fissile material in the history of its nuclear program.

To assess the status of Iran's nuclear program, it is essential to examine the two other requirements of a nuclear weapons capability. Iran largely has satisfied the requirement of delivery systems. Iran has an extensive missile program, which it continues to direct large amounts of resources to improving, in addition to a variety of other delivery systems (more on this in the capabilities section). However, where Iran has the most work to do is in "nuclear weaponization." As Albright explains, nuclear weaponization "includes the mastery of the high explosive triggering system, the molding and machining of high explosives, and the building of a neutron initiator that starts the chain reaction at just the right moment to create a nuclear explosion." According to Albright, there are two possible routes Iran might follow for nuclear weaponization if they choose to move forward in becoming a nuclear capable state. These include:⁶

1. Launching an accelerated effort to achieve a few crude nuclear weapons.
2. Completing a nuclear weapons program with the ability to serially produce annually many warheads suitable for delivery by ballistic missiles.

Iran may choose to follow either path, as each has significant pros and cons. Most notably, there is a trade-off between path 1 and 2 between Tehran's nuclear stockpile, capabilities, and efficacy, and secrecy, international backlash, and speed of achieving weapons capability. The accelerated weaponization program could be completed in six months or less, and would involve activities that could be conducted in smaller, more disisable facilities. This path would offer Iran a far more assured route to establishing itself as a nuclear power while allowing the international community little time to react to its actions before it already possessed a weapons capability. This path is not unprecedented; as Albright explains "It is also the path followed by other programs such as Pakistan's successful effort in the early 1980s and Iraq's in 1990, the latter thwarted by war."⁷ However, this path would undoubtedly leave Iran with a relatively small nuclear arsenal, accompanied by unreliable delivery systems. The second path, however, has some notable challenges. In order to follow the path, Iran would be required to maintain secrecy for an extended period (by many assessments at least two years). The necessary advancements during this period would include a range of production-scale facilities able to produce warheads for ballistic missiles. It is unlikely that Iran would be able to do this without discovery by Israel, the United States, or the international community, in which case the parties would have plenty of time to organize a coordinated response to Iran's push for a nuclear weapons capability. Iran may pursue this route, and perhaps would be

⁶Albright, David. "How Quickly Could Iran Make Nuclear Weapons Today?" Institute for Science and International Security.

⁷Albright, David. "How Quickly Could Iran Make Nuclear Weapons Today?" Institute for Science and International Security.

unmoved by any response against its nuclear advancements short of direct military confrontation, especially given the economic and political support offered to Tehran by Moscow and Beijing.

Although Iran's time-frame for achieving even a crude nuclear bomb may be at least six months, if Iran chooses to pursue path 1, conducting most of the development in secrecy, it is possible the U.S., Israel, and allies may have a far shorter amount of time to react to news of an Iranian push for nuclear weapons. Thus, most strategies espoused by experts, and used by the United States government to this point have focused on deterring Tehran from deciding to build nuclear weapons.

2.3 What policies does the U.S. currently employ, and what have experts recommended?

There are a variety of strategies the U.S. and its partners can employ in trying to prevent Iran from achieving a nuclear weapons capability. Dobbins et al. discuss a useful framework for the spectrum of approaches to deterring Iran. They explain that "Competing U.S. approaches toward Iran might be characterized as engagement versus containment, preemption versus deterrence, and normalization versus regime change. In effect, these are competing archetypes, offering three policy spectrums from which actual courses of action can be chosen."⁸ They explain that engagement would emphasize diplomacy to resolve differences between the West and Iran, although such a policy is unlikely to advance U.S. objectives as long as Tehran is controlled by principlists, and an ascendant Revolutionary Guard. Containment would employ defensive alliances, sanctions, and non-communication to isolate and penalize Iran for its behavior. This approach may constrain Iran's aggressive external behavior, and perhaps prevent progress on its nuclear program. However, the approach would actually work against the goals of reforming domestic politics in Iran, and it might increase support for the Iranian regime, and even potentially for the acquisition of nuclear weapons. The approach would suspend the Middle East and relations with Iran in a precipitous state, without solving some of the key problems at the root of instability, crisis, and conflict. Preemption would include using offensive threats or military action to forestall Iranian weaponization and other undesirable developments. However, preemption would almost certainly increase external sympathy and internal support for the Iranian regime, and could give reason for Tehran to accelerate its nuclear program. Finally, deterrence would employ threats of retaliation against Tehran if it were to cross critical thresholds. Deterrence alone, without other means of de-escalation or greater levels of engagement can lead to increased risks of uncontrolled escalation.⁹ As Dobbins et al. argue, none of the four approaches can or should be taken in isolation,

⁸Dobbins, James, Alireza Nader, Dalia Dassa Kaye, Frederic M. Wehrey, Smith Richardson Foundation, and Rand Corporation, eds. 2011. *Coping with a Nuclearizing Iran*. RAND Corporation Monograph Series, MG-1154-SRF. Santa Monica, CA: Rand Corp. pg. 11.

⁹Dobbins, James, Alireza Nader, Dalia Dassa Kaye, Frederic M. Wehrey, Smith Richardson Foundation, and Rand Corporation, eds. 2011. *Coping with a Nuclearizing Iran*. RAND Corporation Monograph Series, MG-1154-SRF.

as each independently is accompanied by significant detrimental effects and risk. Thus, limited engagement may be used to deescalate tensions with Iran and undermine the regime domestically, containment may be used to blunt the influence and strength of Tehran's proxies, preemption may be threatened only in response to the most undesirable outcomes (although preemption must be used with great caution, if at all), and deterrence strategies can be used to complement containment and engagement by threatening actions against undesirable Iranian initiatives and aggression abroad.

There is almost complete consensus amongst the experts on the subject that a nuclear weapons-capable Iran is a terrifying prospect, one which would be detrimental to security and stability in the broader Middle East. Thus, much of the work done on the subject has considered strategies the U.S. and allies might employ to deter Tehran from making the leap in its nuclear program to having a weapons capability. There are a variety of options available to the United States and allies. Of course, it is possible that Tehran will continue to hedge with its nuclear program, without ever making the leap to a weapons capability. However, if they are to do so, it seems more likely that it would be through strong and effective deterrent policies by the United States. An effective strategy for the U.S. and allies in deterring Tehran from producing nuclear weapons likely involves use of all the political, economic, and military tools at the disposal of the U.S. government and military.

In order to effectively deter Tehran from weapons-capabilities, the U.S. will need to coordinate its effort with other allies to take measures against the threats posed by Tehran. Thus, the U.S. administration would benefit from coordinating with states including France, Germany, the United Kingdom, and other like-minded states to build consensus on the measures each state would be willing to adopt to prevent Tehran from crossing the thresholds that would bring Iran close to nuclear weaponization. Many of these thresholds are generally well-known, as Suzanne Maloney has explained in testimony to Congress.¹⁰ The critical thresholds include enriching uranium to 90% purity, ending all or most access of the IAEA to monitoring or inspecting Iran's nuclear facilities, withdrawing from the NPT, and overt or covert weaponization activities. An effective deterrence strategy would likely necessitate clearly outlining the economic, political, and military consequences imposed by the U.S. and allies upon Iran if it were to cross any of these thresholds.¹¹

Sanctions are a powerful and commonly employed tool in the U.S. and its partners' arsenal. Sanctions may be deployed for both long and short-term purposes. Long-term objectives of sanctions might include bringing Iran fully into compliance with the NPT. Short-term objectives of sanctions would likely be to hinder Iranian progress regarding its nuclear program. Ideally, sanctions would, in the short-term, halt entirely Iranian progress towards weaponization. In order to achieve both long and short-term objectives, sanctions should likely be used as both carrots and

Santa Monica, CA: Rand Corp. pg. 11.

¹⁰Maloney, Suzanne. "Addressing Iran's Evolving Threats to US Interests." Brookings.

¹¹Maloney, Suzanne. "Addressing Iran's Evolving Threats to US Interests." Brookings.

sticks. If Tehran agrees to verification measures, IAEA monitoring, and provides convincing evidence that it has not weaponized, the U.S. and its partners may respond with removal or decreases in sanctions. However, the U.S. and its partners should reserve enough sticks to quickly and comprehensively impose sanctions to facilitate continued progress towards Iranian compliance with the NPT.¹² Many of the United State's European allies tend to view trade and investment with Iran as important tools for positively reinforcing Iranian cooperation with Western de-escalatory interests. Thus, many of our allies have been hesitant to employ sanctions and international political pressure against Iran. In fact, until recently, many of the United State's European allies have offered certain export credits and trade incentives in order to promote economic cooperation with Tehran. However, each of the United State's European allies have also played a crucial role in negotiations with Tehran to constrain its nuclear program at various points in time. And, any further progress towards constraining Iran's nuclear program and deterring Tehran from a weapons-capability will require extensive coercive measures by European allies either through imposing sanctions or exerting international pressure on Tehran. The same diplomatic infrastructure that Washington built to impose significant sanctions against Russia in response to its invasion of Ukraine can be used to coordinate a sanctions regime against Iran. Unfortunately, as Tehran has pivoted to the East, developing important economic relationships with Russia and China, the West's ability to cripple Iran's economy has diminished. Thus, although cooperation with Russia is likely largely out of the question, the West may be able to push China to cooperate with the West in preventing Tehran from proceeding with nuclear weaponization. China has historically been a free-rider in nuclear diplomacy with Tehran, so it seems unlikely China would be willing to cooperate with the U.S. on any such initiatives, especially as tensions reach new heights between Washington and Tehran. However, due to China's status as a major importer of Iranian oil, if the West wishes to impact Iran's economy in any significant way, cooperation from Beijing may be necessary.

There are critical military initiatives that can be used to signal the resolve of the United States concerning Tehran's nuclear program. Sanctions and international political pressure can be employed to hinder Iran's progress concerning its nuclear program, debilitate its economy, and present a strong allied consensus against further progress by Iran towards nuclear weaponization. Military initiatives can demonstrate the credibility of U.S. and allies response to Tehran crossing certain critical thresholds towards nuclear weaponization. By increasing military exercises in coordination with European allies and Israel in the Middle East, the U.S. can signal coalition capability to undertake operations necessary to strike Iran's nuclear facilities, either waylaying or destroying all together Tehran's nuclear program. Some of these military initiatives might include a high level of U.S. force deployment in the Persian Gulf, bolstering and facilitating increased security plan-

¹²Dobbins, James, Alireza Nader, Dalia Dassa Kaye, Frederic M. Wehrey, Smith Richardson Foundation, and Rand Corporation, eds. 2011. *Coping with a Nuclearizing Iran*. RAND Corporation Monograph Series, MG-1154-SRF. Santa Monica, CA: Rand Corp. pg. 16.

ning and cooperation with Israel and other Gulf states, and investing in infrastructure necessary to strengthen regional air defense systems. All of these initiatives would not only ensure that the U.S. and allies are more prepared to respond to any Iranian threats, but also that they are more willing to follow through on commitments to preventing Tehran from achieving a weapons capability.¹³

2.4 Deterrence Failure: Why might we see a nuclear weapons-capable Iran?

The American track record on Iran has been modest at best. Washington and its allies have been able to slow Tehran's nuclear program throughout its 30-year history, have prevented achievement of a nuclear weapons capability, and have blunted some of Iran's proxies in the Middle East. However, there have been few meaningful breakthroughs or sustained agreements preventing critical aspects of Tehran's nuclear program. Today, the Iranian government's capacity for aggression in the Middle East is at its highest, and its nuclear program is more advanced than ever before. Iran is closer than ever to a nuclear weapons capability, and the prospects for meaningful deterrence against a push from Tehran for a weapons capability are limited.

Would the U.S. stop Iran if it chose to pursue a nuclear weapons capability? What is U.S. policy? What has the U.S. said? Would the U.S. launch a preventive/preemptive strike against Iran if it chose to weaponize? Would the U.S. go to war with Iran to stop it from having nuclear weapons? Would the U.S. even support an Israeli preventive strike or war against Iran to prevent Tehran from achieving a nuclear weapons capability?

What is Israeli policy if Iran chooses to pursue nuclear weapons? Would Israel launch a preventive strike/ initiate preventive war to stop Iran from having nuclear weapons? If they chose to do so, would such an initiative or strike be successful?

There are a few reasons why Israel would not initiate war against Iran, even in the event Tehran chooses to make the final push more nuclear weapons:

- Israel's current war with Hamas in Gaza has depleted Israel of the necessary munitions, resources, and systems to engage with Iran. Iran will not necessarily pursue a nuclear weapons capability while Israel is engaged in war against Hamas. However, if Iran were to pursue a nuclear weapons capability while Israel is in conflict with Hamas or any of Iran's proxies, Israel would likely be depleted of the systems and resources necessary to engage Iran directly. Israel is unable to fight a multi-front war against Iran's proxies, let alone direct conflict with Iran.
- Israel would be counting on the support of the United States if they were to strike Iranian nuclear targets, support which is, at best for Israel, uncertain to be offered. The U.S. has tempered its more bellicose rhetoric of a preemptive option, and largely lacks the support

¹³Maloney, Suzanne. "Addressing Iran's Evolving Threats to US Interests." Brookings.

of its European allies for any such action. “Europe, for its part, was reluctant to embrace George W. Bush’s early emphasis on preemption as a response to nuclear proliferation. As Bush in his second term and then Barack Obama tempered this bellicose rhetoric, the major European powers began to align themselves more closely with Washington’s efforts to employ mounting sanctions to stem the Iranian nuclear program. Russia and China have done likewise, albeit more cautiously.”¹⁴ Along with other historical precedent, it is highly unlikely that the United States will support or initiate a preventive war. The United States is therefore unlikely to support offensive preventive measures by Israel against a developing or nascent Iranian nuclear weapons capability.

- Israel is facing immense pressure from the international order against any further aggression. Israel is worse-positioned currently and in the near future than ever before in its history to take action against another state or non-state entity militarily, in terms of international support or the lack thereof. Israel would face significant backlash for any military action in which it is perceived as the aggressor. Thus, even if Israel was acting on the basis of retaliation against Iran for the actions of its proxies, or for Iran’s nuclear ambitions, Israel would have to persist through significant opposition, and a lack of international support.
- Israel might not possess the capabilities to strike Iran’s nuclear delivery systems and critical infrastructure. The hardening, concealment, and redundancy of Iran’s nuclear facilities might mean that even the most successful version of an Israeli attack would be unable to prevent, delay, or even hinder Iran’s nuclear weapons program, and achievement of a weapons capability. Limitations of a successful Israeli preventive strike include the structure, size, and location of Iran’s conventional and nuclear forces, but also the size and precision capabilities of Israel’s offensive weapons systems.
- Even a 100% successful initial counter-force strike by Israel against Iran’s nuclear infrastructure might lead to undesirable outcomes. Any strike of the kind would lead to a long, bloody, and costly hot war with Iran and its proxies. This war in itself could threaten the existence of Israel, and could therefore render a preventive attack/war against Iran illogical.

3 Capabilities Assessment

To effectively analyze the implications of a nuclear capable Iran for the Middle East, and U.S. prospects for deterrence and deescalation under various conditions, it is important to describe the military capabilities and strategic doctrine of Iran, its proxies, and Israel. The following sections

¹⁴Dobbins, James, Alireza Nader, Dalia Dassa Kaye, Frederic M. Wehrey, Smith Richardson Foundation, and Rand Corporation, eds. 2011. *Coping with a Nuclearizing Iran*. RAND Corporation Monograph Series, MG-1154-SRF. Santa Monica, CA: Rand Corp. pg. 8.

will present a technical account of the military forces and posture of each of Iran, Iran's proxy forces, and Israel.

3.1 Iran

3.1.1 Iran's Military Doctrine

Iran's military and strategic doctrine is complex, informed by a history of conflict with regional and international powers. As Michael Connell discusses, the Iranian military doctrine constitutes a hybrid of Western military concepts combined with "martyrdom and revolutionary zeal."¹⁵ Iran's military forces have been structured to confront technologically superior adversaries (the United States and Israel). Iran relies on a "deterrence-based model of attrition warfare." This model, while not necessarily ensuring Iranian victory in any kind of conventional or nuclear conflict, emphasizes imposing high costs on the adversary to inhibit their resolve and willingness to fight or engage in conflict. Asymmetric warfare is also central to Iran's military doctrine. Thus, Iran's military structure plays to its comparative strengths in the Middle East: manpower, strategic depth, and a willingness to sustain losses and high costs. Similarly, Iran's strategic doctrine targets its potential adversaries weaknesses and dependencies: risk-aversion, casualty sensitivity, high reliance on technology, and dependence on limited regional basing access.

"Iran's armed forces consist of two separate, parallel militaries—the Artesh, or regular forces, and the IRGC. The Artesh, which long predates the revolution, focuses on defense against external threats; the mission of the IRGC, formed from armed militias during the revolution, is to defend the regime and its Islamic system of government from any threat, foreign or domestic."¹⁶ "The Artesh primarily focuses on defending Iran's borders from external threats. It consists of ground, naval, air, and air defense components, which in total number about 420,000 personnel."¹⁷ The "Islamic Revolutionary Guard Corps Supreme Leader Khomeini established the IRGC, also known as the Sepah (corps) or Pasdaran (guard), shortly after the Islamic Revolution. It has a broader mission to defend the Iranian revolution from any foreign or domestic threat. The IRGC designated as an FTO by the United States—consists of ground, naval, aerospace, and unconventional components, which in total number about 190,000 personnel. Including the estimated active personnel from Iran's paramilitary reserve force, the Basij, the IRGC numbers roughly 640,000 personnel."¹⁸ "Iran's constitution designates the supreme leader, not the president, as commander in chief of the armed forces, with the power to exercise military C2, declare war and peace, and approve military operations. Khamenei traditionally issues orders through the AFGS and KCHQ, which oversee and coordinate between the IRGC and Artesh, but sometimes bypasses these organizations to give

¹⁵Connell, Michael. 2010. "Iran's Military Doctrine — The Iran Primer." October 11, 2010.

¹⁶"Defense Intelligence Agency Releases Report: Iran Military Power." pg. 10.

¹⁷"Defense Intelligence Agency Releases Report: Iran Military Power." pg. 25.

¹⁸"Defense Intelligence Agency Releases Report: Iran Military Power." pg. 25.

orders directly to lower-level commanders.”¹⁹

Iran wishes to avoid direct confrontation with the United States or Israel. A conventional or nuclear conflict with either state would cause severe damage to the Iranian state, and would constitute a major threat to the survival of the Iranian regime. Thus, at a strategic level, Iran’s ground, naval, and air forces are primarily defensive in nature. Iran’s ground forces are structured to take advantage of Iran’s strategic depth and geographic characteristics, making a ground invasion a difficult prospect for even the world’s strongest military’s. Iran’s maritime strategy is designed to deter conflict. If deterrence fails, the IRGC-N and IRIN would seek to impose significant costs on the adversary through relatively low-cost means. Operationally, this involves the integration of sea, land, and air defenses to create a layered defense, featuring a web of threats with increased density close to the Iranian coastline. Iran’s greatest weaknesses and strengths are through the air. Iran’s air force is weak, operating with outdated planes and systems. However, Iran’s greatest strength is in its missile and air defense capabilities. According to Alterman et al. “missiles and rockets serve Iran’s larger defense posture by holding at risk its neighbors and U.S. forward forces, imposing significant costs for defense investment for its rivals, and providing a hedge for future nuclear capability”²⁰ “While Iran could not defeat the combined forces of the United States, the GCC, and Israel in open warfare, missiles and rockets could make such a conflict costly enough to give these powers pause in directly confronting Iran’s regime, thereby increasing its freedom of action. To prevent this dynamic, the United States and Iran’s neighbors will need to continue to counter Iran’s efforts, including with missile defense sensors and interceptors.”²¹ “Together, these trends mean that the freedom of access and control that the United States has for decades enjoyed in the Gulf will become more difficult to maintain. Iran’s A2AD and retaliatory capabilities are likely to evolve in quantity and quality to a point where it may even deter the United States or other countries from taking action against its nuclear program in the event of breakout, thereby undermining regional stability, U.S. credibility, and broad interest in maintaining a diplomatic solution to the nuclear issue.”²² While Iran’s air forces and missile capabilities are largely defensive, some of the systems Iran possesses can be easily used for offensive purposes, or be converted with little difficulty to constitute an offensive capability within the Middle East.

One of the most important elements of Iran’s regional and international strategic doctrine is its deployment of and support for proxy militias, forming what Iran refers to as the “Axis of Resistance.” Over the past decades, with limited effective mitigation by regional states or international coalitions, Tehran has assembled an extensive and powerful network of proxy forces in the Middle

¹⁹“Defense Intelligence Agency Releases Report: Iran Military Power.” pg. 26.

²⁰Alterman, Jon B., Michael Connell, Michael Eisenstadt, Farideh Farhi, Thomas Karako, J. Matthew McInnis, Hijab Shah, et al. 2017. “Deterring Iran after the Nuclear Deal,” pg. 64.

²¹Alterman, Jon B., Michael Connell, Michael Eisenstadt, Farideh Farhi, Thomas Karako, J. Matthew McInnis, Hijab Shah, et al. 2017. “Deterring Iran after the Nuclear Deal,” pg. 65.

²²Alterman, Jon B., Michael Connell, Michael Eisenstadt, Farideh Farhi, Thomas Karako, J. Matthew McInnis, Hijab Shah, et al. 2017. “Deterring Iran after the Nuclear Deal,” pg. 65.

East. While each proxy is beholden to the Iranian regime to differing extents, the overlapping interests of the proxy leaderships and Tehran, along with massive economic and military support from Iran has enabled the Islamic Republic to wield significant influence in the Middle East through its proxies. Tehran has spent decades cultivating ties with proxy militias in the greater Middle East by providing support, funding, weapons, and training to its proxies. Iran's proxy forces are trained, organized, and directed by "The executor of Iranian proxy policies, the Islamic Revolutionary Guard Corps Qods Force (IRGC-QF)." The IRGC was created by Supreme Leader Ayatollah Ruhollah Khomeini, and has been responsible for Tehran's relationship with its proxies.²³ Iran suffers from limited abilities to project power abroad, which is where its proxies offer critical advantages. Tehran uses its proxies for a variety of purposes. Proxy forces are effective at promoting Iran's ideological and political influence. Iran's proxies are a crucial part of its deterrence strategy. One such method of deterrence is through the threat of retaliation, that Iran's proxies can impose significant costs and casualties on Iran's adversaries in response to direct military action against Iran or its interests. Retaliation by Iran's proxies also allow for a degree of separation or plausible deniability for Tehran, allowing the Iranian regime to manage escalation by its adversaries. The proximity of Tehran's proxies to U.S. forward deployed forces, and Israel's territory allows Iran to balance the deterrence equation through asymmetric escalation and targeting of U.S. and Israeli entities in the region. Iran's proxies have a passive deterrent effect, where through continued operations in Iran's sphere, international powers are discouraged from engaging in the region or interfering with Iranian interests due to the constant threat of proxy aggression and violence. While Iran does not wish to provoke or engage with the U.S. or Israel in direct military conflict, it has enabled numerous attacks and acts of aggression and violence by its proxies against Israel and U.S. forces in the greater Middle East.

3.1.2 Iran's Military Capabilities

Iran's missile force is the largest and most diverse in Asia. The Islamic Revolutionary Guard Corps Strategic Aerospace Force (IRGCSAF) operates the majority of Iran's ballistic missile forces through the Al-Ghadir Missile Command. As Alterman et al. discuss, Iran's missile force primarily "comprises a mix of medium-range ballistic missiles (MRBMs), short-range ballistic missiles (SRBMs) and rockets, land-attack cruise missiles, antiship missiles, and air defenses. Iran has made major progress in producing these systems domestically, and has taken steps to increase the mobility and survivability of these assets, including the construction of underground missile bases and hardened launch silos."²⁴ Iran's Close Range Ballistic Missiles (CRBMs) and Short Range Ballistic Missiles (SRBMs) include the Fateh 110, with a range of roughly 300 km, the Fateh

²³Alterman, Jon B., Michael Connell, Michael Eisenstadt, Farideh Farhi, Thomas Karako, J. Matthew McInnis, Hijab Shah, et al. 2017. "Deterring Iran after the Nuclear Deal," pg. 37.

²⁴Alterman, Jon B., Michael Connell, Michael Eisenstadt, Farideh Farhi, Thomas Karako, J. Matthew McInnis, Hijab Shah, et al. 2017. "Deterring Iran after the Nuclear Deal."

313 (range of roughly 500 km), and the Zolfaghar SRBM (approximately 700 km). Iran also employs the Khalij Fars, Hormuz 1, Hormuz 2, which are all close range missile systems capable of targeting ships, with ranges of approximately 300 km. The most notable of Iran's rockets is the Fajr series, which are cheap but inaccurate and unreliable rockets. Iran's Medium Range Ballistic Missiles (MRBMs) most notably include the Shabab 3 series, Shejjil, and Khorramshahr. Iran's MRBMs purportedly have a range of about 2000 km, although they currently lack significant precision capabilities. Iran's newer Land Attack Cruise Missiles (LACMs) include the Meshkat and Soumar systems which have a reported range of about 2000 km. Critically, these LACM systems may have a greater precision strike capability than most of the other systems in Iran's missile forces. Intelligence estimates from the past few years suggest Iran possesses over 800 MRBMs and SRBMs. While Iran's ballistic missile arsenal still likely lacks accuracy necessary for strikes on smaller, mobile, or hardened military targets, the large quantity of MRBMs makes Iran's ballistic missile force a severe threat. Iran's missile forces have the ability to pose significant risk to softer targets, including urban areas, which provides Iran with leverage over many other regional actors, including Israel.²⁵ Iran continues to update its existing systems, and develop new systems with great precision capabilities and range. Iran has also began work on space launch vehicles (SLVs), including the Simorgh. Development of such a system would shorten the timeline to produce an ICBM, if Tehran decided to develop one, as the systems use similar technologies.²⁶

“Medium-range ballistic missiles, for example, afford Iran the ability to conduct salvo attacks against larger military installations or urban centers in the Gulf Cooperation Council (GCC) countries, Israel, and southwestern Europe. Shorter range anti-ship missiles and growing air defense capabilities also give Iran a modest anti-access/ area-denial (A2AD) capability, which can threaten U.S. maritime and air forces and thereby undermine their deterrent value. As observed by General Lloyd Austin III, then U.S. Central Command (CENTCOM) commander, Iran's ambitions for regional hegemony are aided by their concert and employment of “various anti-access and area denial (A2/AD) capabilities, theater ballistic missile and cyber capabilities, [and] aggressive maritime activities” and by the Quds Force and other proxies.”²⁷ While Iran's SRBMs and MRBMs are not explicitly advertised as built for delivering a nuclear payload, most of these systems are inherently dual purpose. The large majority of Iran's ballistic missile systems purportedly can carry a warhead of over 500 kilograms, which would hypothetically allow the systems to carry and deliver nuclear warheads.²⁸ Iran also possesses a large arsenal of unmanned aerial vehicles (UAVs). “UAVs are Iran's most rapidly advancing air capability. Iran uses these versatile platforms for a

²⁵ Alterman, Jon B., Michael Connell, Michael Eisenstadt, Farideh Farhi, Thomas Karako, J. Matthew McInnis, Hijab Shah, et al. 2017. “Deterring Iran after the Nuclear Deal,” pg. 66.

²⁶ Annual Threat Assessment of the U.S. Intelligence Community.” 2024.

²⁷ Alterman, Jon B., Michael Connell, Michael Eisenstadt, Farideh Farhi, Thomas Karako, J. Matthew McInnis, Hijab Shah, et al. 2017. “Deterring Iran after the Nuclear Deal,” pg. 64-65.

²⁸ Alterman, Jon B., Michael Connell, Michael Eisenstadt, Farideh Farhi, Thomas Karako, J. Matthew McInnis, Hijab Shah, et al. 2017. “Deterring Iran after the Nuclear Deal,” pg. 100.

variety of missions, including ISR and air-to-ground strikes. The IRGCASF is the primary operator of Iran's growing fleet of UAVs, although most Iranian military services employ them."²⁹ "Iran has steadily expanded its domestic UAV inventory by developing and producing a wide range of lethal and nonlethal UAV platforms. Iran has reverse-engineered many of these systems based on captured Western UAVs. Some of Iran's newer UAV platforms, such as the Shahed-129 and the Mohajer-6, can be armed and are capable of conducting precision air-to-ground strikes with small guided munitions. However, despite advances in its UAV manufacturing capabilities, Iran remains reliant on Western manufactured engines and components to support its UAV production. Iran is developing a domestic UAV engine but is struggling with quality issues."³⁰

The Islamic Republic of Iran Air Defense Force (IRIADF) maintains and operates most of the air defense systems of Iran. The Khatemolaniah Air Defense Headquarters (KADHQ) oversees Iran's national-level air defenses across IRIADF and IRGCASF. KADHQ controls air defense command and control (C2), air surveillance radars, SAM systems. KADHQ C2 is centralized during peacetime, at the national Air Defense Operations Center. However, C2 can be decentralized during crisis or conflict, transferring decision-making authority down to a network of fixed and mobile sector operations centers (SOCs), which manage air defense operations within their areas of responsibility, while coordinating with other SOCs. The most notable of Iran's air defense systems is the SA-20c (also known as the Russian S-300), acquired by Iran from Russia, will most likely be used to protect Iran's most critical infrastructure (nuclear sites, Tehran). The SA-20c air defense system is highly mobile, and designed to defend against aircraft, ballistic missiles, and cruise missiles. Iran's air defense systems also rely on various secondary defenses. These include a mix of U.S., Russian, Chinese legacy systems, including the long-range SA-5, medium range I-HAWK and CSA-1, and the short range SA-15 and Rapier. Iran is also currently developing and implementing the long range Bavar-373 SAM system, which is developed and produced largely domestically. Iran claims that this system is more advanced than Russian S-300, and will eventually supplement then replace the system, providing a greater degree of air defense. Iran also continues to update systems to improve legacy surface to air missiles (SAMs) and other air defenses, including the medium range Mersad system, which improves the tracking and engagement range of the U.S. I-HAWK SAM. Iran is also in the process of upgrading legacy C2 systems to modern, software-based systems to further improve the reliability and expediency of its air defense systems.³¹

Iran's greatest weakness is its air force, as much of the country's aircraft date from the era of Shah Mohammed Reza Pahlavi, who led Iran from 1941 to 1979. However, Iran has purchased some newer aircraft from Russia in the 1990s, and has started taking delivery of advanced trainer

²⁹"Defense Intelligence Agency Releases Report: Iran Military Power." pg. 67.

³⁰"Defense Intelligence Agency Releases Report: Iran Military Power." pg. 87.

³¹"Defense Intelligence Agency Releases Report: Iran Military Power."

aircraft and probably will seek to acquire new conventional weapon systems, such as advanced fighter aircraft and helicopters. The IRIAF operates the majority of Iran's air force, primarily through 11 major fighter bases. The most notable of Iran's aircraft include legacy U.S., Russian, and Chinese aircraft. U.S. aircraft include the F-14 Tomcat, F-4 Phantom II, and F-5 Tiger II (the F-4 is the primary attack aircraft). Iran also uses the Russian MiG-29 Fulcrum and Su-24 Fencer, and the Chinese F-7 Airguard.³²

3.2 Iran Proxies

Iran's network of proxies throughout the Middle East is extensive. Tehran, facilitated by the IRGC has close ties to proxy militias covering the Middle East. These proxy forces include Hizballah (Lebanon/Syria/Iraq/Yemen), Iraqi Shia Militias (Iraq), Huthis (Yemen), Palestinian Militants (Palestinian Territories) most notably HAMAS, Taliban (Afghanistan), Bahraini Militants (Bahrain), and Shia Foreign Fighters (Syria). Hizbollah poses the greatest threat to Israel, although HAMAS too has the ability to conduct costly and damaging attacks against Israel, as demonstrated by the ongoing war. The Huthis and Hizbollah pose the greatest ongoing threat against U.S. personnel and infrastructure. Iran's other proxy forces are not to be discounted, as they each receive deadly military systems and training from Iran. However, for the sake of this analysis it is sufficient to describe the capabilities of the strongest of Tehran's proxies.

"The level of support and types of systems Iran has provided for nonstate actors is really unprecedented in terms of drones, ballistic missiles and cruise missiles," said Fabian Hinz, an expert on Iran's military at the International Institute for Strategic Studies in Berlin."³³ Iran has provided its proxies with the weapons systems necessary to produce scores of "militia, rocket, missile, and UAV attacks against U.S. forces in Iraq and Syria; Hizballah exchanges of fire with Israel on the north border with Lebanon; and Huthi missile and UAV attacks, both on Israel directly and on international commercial shipping transiting the Red Sea."³⁴

Since October 2023, Iran has encouraged and enabled its various proxies and partners—including Hizballah, Iranian-backed groups in Iraq and Syria, and the Huthis in Yemen—to conduct strikes against Israeli or U.S. interests in the region. Hizballah is calibrating this pressure on Israel from the north while trying to avoid a broader war that would devastate Hizballah and Lebanon. Hizballah's leadership, though, probably will consider a range of retaliatory options depending on Israel's actions in Lebanon during the upcoming year. "In Iraq, Iranian-aligned militias almost certainly will continue attacks against U.S. forces in Iraq and Syria. The Huthi's continued ballistic missile, cruise missile, and UAV attacks against merchant vessels transiting the Red Sea, which are disrupting international shipping, and on Israel create a real risk of broader escalation."³⁵

³²"Defense Intelligence Agency Releases Report: Iran Military Power."

³³Fassihi, Farnaz. 2024. "A Look at Iran's Military Capabilities." The New York Times, April 12, 2024.

³⁴Annual Threat Assessment of the U.S. Intelligence Community." 2024. pg. 18-19.

³⁵Annual Threat Assessment of the U.S. Intelligence Community." 2024. pg. 24-25.

3.2.1 Hezbollah

Hezbollah is a Shiite Islamist political and militant group based in Lebanon. Formed in the early 1980s, Hezbollah's primary goals include resisting Israeli presence in Lebanon and advocating for the establishment of an Islamic state. The group is recognized for its military capabilities and political influence within Lebanon. Hezbollah is designated as a terrorist organization by several countries, including the United States and many members of the European Union. However, Hezbollah also enjoys substantial support within Lebanon and the wider Arab and Muslim worlds, where it is seen as a resistance movement against Israeli aggression and Western imperialism. Hezbollah, of all of Iran's proxies in the Middle East, constitutes the greatest threat to the State of Israel, in large part due to its sophisticated weapons capabilities supplied by Iran. Hezbollah, over the past decades, has become the world's most heavily armed non-state actor. It possesses an arsenal which both qualitatively and quantitatively dwarfs its previous capabilities, and those of most other actors in the Middle East.

Hezbollah possesses an arsenal estimated at somewhere between 150,000 - 200,000 projectiles. These include "unguided artillery rockets, as well as ballistic, anti-air, anti-tank, anti-ship missiles, mortars, and drones." The bulk of Hezbollah's arsenal consists of short-range and unguided projectiles, including a variety of rockets with ranges from 4 - 40 km, and able to carry warheads with between 10 - 20 kg of high explosives or sub-munitions. "According to a 2023 study by Tel Aviv University's Institute for National Security Studies, Hezbollah has about 40,000 rockets within the Katyusha variants, complemented by about 80,000 slightly longer range Fajr-3 (45km, 45kg) and Fajr-5 (75 km, 90kg), Raad 2 (65km) and Raad-3 (70km, 50kg), and Khaibar-1/M-302 (100km, 150kg). If fired from near Metula, this part of Hezbollah's arsenal alone could reach halfway between Haifa and Tel Aviv, or close to Herzliya if fired from near Naqoura, at the southernmost possible point. Additionally, during the current conflict, Hezbollah revealed that it also possesses the Iranian-made Falaq-1 rockets, using them for the first time in strikes against Israel beginning on January 26, 2024. The Falaq-1 is an unguided surface-to-surface missile, capable of carrying a 50kg warhead, with a maximum range of 10km. It is launched from platforms mounted to the rear of 4x4 light jeeps, which have six Falaq-1 rocket launchers in a ready-to-launch position." "Israel's primary countermeasure to this tier of rockets is the Iron Dome. Different estimates exist of this system's accuracy, but its median success rate is approximately 90 percent — which means that 10 percent of the number of rockets fired into Israel will likely succeed in penetrating northern Israel at various depths – which is 100-150 rockets at the low range, and 300-400 rockets at the higher range. But that is the ideal world scenario, which assumes 1) Israel has infinite funds and infinite Tamir Interceptors for its Iron Dome batteries, 2) Iron Dome performs optimally, and 3) Hezbollah doesn't employ tactics to throw off Iron Dome's radar. The price of each interceptor and Israel's Tamir stockpile are classified. But most estimates suggest that, given the expected rate

and quantity of Hezbollah's fire, this stockpile would 'run out within mere days of the onset of fighting, leaving Israel exposed to thousands of rockets and missiles at all hours of the day, without any effective active defense, and certainly not a hermetic one.'"³⁶

"Hezbollah also possesses an assortment of drones – including Mirsad-1 (200km range), Ayoub, a Shahed-129 derivative (1600+ km range), and potential access to other Iranian-made drones like the Karrar, Mohajer, and Samed. Hezbollah also possesses loitering munitions, which it has used in recent attacks on Israel. According to strategic analyses, launching a large number of drones could very likely overwhelm Iron Dome's capabilities. While the exact number of drones in Hezbollah's possessions is unknown, one Israeli think tank estimates that the group possesses as many as 2000 UAVs – though how it came about this estimate is unknown." Iron Dome can theoretically engage drones, but its effectiveness has not been tested against swarms of drones, and their flat trajectory makes it harder for the missile defense system to be as effective with them as with rockets. This could allow Hezbollah to penetrate many kilometers into the Israeli Home Front and target critical civilian and military infrastructure. This drone arsenal would pose a sufficient challenge in and of itself to Israel, particularly if Hezbollah launches swarms of drones or loitering munitions. But Hezbollah could also combine drone swarms with rocket barrages, throwing off Iron Dome's radar and targeting system, and allowing more of the group's projectiles to hit targets within Israel. Credible assessments suggest that by using a combination of rocket barrages, "loitering munitions, [and] armed drones," Hezbollah could "bring about the collapse of [Israel's] missile defense system, completely exposing Israel to rocket fire directed at it."³⁷

Hezbollah's Anti Tank Guided Missiles (ATGM) also constitute a significant part of its offensive military capabilities. "Since beginning its attacks against Israel on October 8, Hezbollah has also demonstrated perfecting a different use for them as terror weapons. Hezbollah has been targeting soldiers and civilian and military infrastructure along the border. This use of ATGMs as a precision weapon, essentially to snipe at more vulnerable Israeli targets, especially civilian ones, has been described as "unprecedented in Israel and perhaps the world." Critically, when used in this manner, Israel has no defense against ATGMs." "But the challenge posed by most of Hezbollah's short-range arsenal is its mobility. Most of the group's projectiles in this category – be they rockets or ATGMs – are typically launched from highly mobile platforms, like trucks or tripods. Their operators will be able to strike and disappear before they can be located and destroyed by Israeli jets. That is why Israel, which relied heavily on its air force during the 2006 Lebanon War, failed to silence Hezbollah's rocket fire on northern Israel until the ceasefire went into effect on August 12."³⁸

³⁶Daoud, David. 2024. "Breaking down Hezbollah's Rocket Strategy: The Short-Range Threat — FDD's Long War Journal." February 26, 2024.

³⁷Daoud, David. 2024. "Breaking down Hezbollah's Rocket Strategy: The Short-Range Threat — FDD's Long War Journal." February 26, 2024.

³⁸Daoud, David. 2024. "Breaking down Hezbollah's Rocket Strategy: The Short-Range Threat — FDD's Long War Journal." February 26, 2024.

3.2.2 HAMAS

Hamas, or the Islamic Resistance Movement, was founded in 1987 during the First Intifada. Hamas is an Islamist Palestinian nationalist organization which seeks to establish an Islamic state in historic Palestine, including present-day Israel, and is committed to the destruction of Israel. Its ideology is rooted in Sunni Islamism and Palestinian nationalism. Although Hamas is Sunni and Iran is predominantly Shiite, they have maintained a pragmatic relationship based on their shared opposition to Israel. Iran has provided Hamas with financial support, weapons, and training. Despite occasional tensions, Iran and Hamas collaborate to counterbalance Israeli and Western influence in the region. Hamas has demonstrated throughout its current war against Israel the capability to pose a serious threat to the lives of Israeli citizens, and to damage population centers and critical infrastructure. Hamas has far fewer resources than some of Iran's other proxies, in large part due to somewhat effective isolation strategies by Israel. However, Hamas still possesses some Anti-Tank Guided Missiles, and more importantly large numbers of small- to mid-range rockets. These rockets are unreliable and inaccurate, but are very inexpensive to produce. It now has a military arsenal that is believed to include thousands of rockets, including long-range missiles and drones. Hamas officials have said their fighters also possess a range of bombs and mortars.

3.2.3 Houthis

The Houthis are a Shiite insurgent group from northern Yemen. The movement began in the early 1990s as a revivalist organization promoting Zaydi Shia Islam but transformed into a militant group by the early 2000s. The Houthis advocate for greater autonomy for the Zaydi-majority regions of Yemen, opposition to the Yemeni government's corruption, and resistance to Saudi and U.S. influence in Yemen. Their slogan, "God is Great, Death to America, Death to Israel, Curse on the Jews, Victory to Islam," reflects their anti-imperialist and anti-Israel stance. The Houthis, Iran's proxy in Yemen, continues to attack both Israeli and U.S. ships and critical shipments in the Red Sea, the southern region of Israel, and the Strait of Hormuz. The Houthis do so through a diverse arsenal of Short Range Ballistic Missiles (SRBMs), Surface-to-Air Missiles (SAMs), Unmanned Aerial Vehicles (UAVs), Anti-Ship Cruise Missiles (ASCMs), and Naval Mines.

3.3 Israel

3.3.1 Israel's Military Doctrine

The Israeli Defense Force (IDF) strategic doctrine is the conceptual framework for Israel's military operations. It is, as such, based on vital national interests and basic assumptions concerning Israel's national security. Israel's strategic doctrine is constructed in the face of a variety of threats, including those from states - distant (Iran) and nearby (Lebanon), failed states and states in a pro-

cess of disintegration (Syria), sub-state organizations (Hezbollah, Hamas, Huthis), and terrorist organizations without links to a particular state or community (Islamic Jihad, Palestinian Islamic Jihad, ISIS, and others). There are a few central principles of Israel's security doctrine which relate to its military conduct, operations, and systems. These include deterrence, early warning, defense, and decision, which are expressed in the following principles. Israel relies on a defensive security strategy to deter attacks on the homeland, and maintain Israel's territorial integrity, by neutralizing threats as they arise. Israel also relies on certain offensive military concepts, as it assumes that a purely defensive strategic posture is insufficient to ensure the survival of the State while defeating threatening adversaries. Thus, Israel uses force in an offensive capacity to achieve military successes and demonstrating resolve in confronting threats against the State of Israel. Israel also relies extensively on strategic cooperation with allies, most notably through its close diplomatic and military relations to the United States. Due to Israel's lack of strategic depth, disadvantages in territory, population size, and hazardous geographic location, Israel also strive to maintain a Qualitative Military Edge (QME) over its adversaries. This includes maintaining advantages over its adversaries in human quality, advanced technological capabilities (weaponry) and various types of intelligence. While Israel generally wishes to avoid direct conflict, especially with hostile states, if confrontation is forced upon Israel or deterrence fails, its strategic doctrine calls for swift and decisive engagement through whatever means necessary to neutralize the threat, and ensure the survival of the state.

3.3.2 Israel's Military Capabilities

While Israel has never publicly acknowledged or claimed to possess nuclear weapons, it is believed to possess roughly 90 nuclear warheads. According to some sources, this is the lower end of the estimate range, and Israel may possess as many as 300 nuclear warheads. According to estimates, about 30 nuclear weapons are gravity bombs. These gravity bombs would be delivered by F-16I or F-15 aircraft. Israel is believed to store its warheads separately from deployed launchers during peacetime, however locations of the storage sites are unknown. Much of Israel's fissile material is plutonium, which is produced at the Negev Research Center (NRC) near Dimona, Israel. Israel has many advanced aircraft, many of which are supplied to Israel by the United States. These include the very advanced F-35 aircraft, of which Israel has about 50. These aircraft, among others, could be used to deliver nuclear weapons, although it remains unclear whether or not Israel plans to use certain aircraft for nuclear weapons delivery.

Israel has one of the most technologically advanced missile arsenals in the Middle east. Aided by foreign assistance and collaboration over the past decades, Israel produces many cruise and ballistic missile systems domestically. Israel's missile forces consist primarily of shorter range, tactical systems, but Israel also possesses a contingent of long-range ballistic missiles to be used

for strategic deterrent purposes. Israel's short range missile forces notably include the Popeye Air Launched Cruise Missile (ALCM) with a range of 75-100 km, the LORA Short Range Ballistic Missile (280 km), the Delilah Land Attack Cruise Missile (LACM) (250-300 km), and the Gabriel and Harpoon Anti-Ship Cruise Missiles (ASCM), with a range of up to 400 and 240 km, respectively. Israel's medium and long range ballistic missile forces include land- and sea- based missiles. The most notable of Israel's ballistic missiles are the Jericho class ballistic missiles. The Jericho II Medium Range Ballistic Missile (MRBM) has a range of between 1,500 - 3,500 km, and the Jericho III Intermediate Range Ballistic Missile (IRBM) has a range between 4,800 - 6,500 km.³⁹ According to intelligence, 50 nuclear warheads are assigned for delivery via Jericho ballistic missiles. These systems are located with their mobile transporter-erector-launchers (TELs), in caves or bunkers at the Sdot Micha Airbase near Zekharia, (25 km west of Jerusalem). It is likely that each of 23 bunkers can store two launchers. These ballistic missiles include a mix of the solid-fuel two-stage Jericho II MRBM, and the newer three-stage Jericho III. At this stage, it is possible that the Jericho III may have completely replaced Jericho II. Israel also possesses 5 German Dolphin-class diesel-electric submarines (the Dolphin-I and Dolphin-II) based in Haifa, and operated in the Mediterranean. These submarines allow for the delivery of ballistic missiles with the capacity to carry nuclear warheads from sea-launched platforms. Some, or all of the submarines are equipped to launch the Popeye cruise missile which is capable of delivering nuclear warheads.

Israel's air-defense systems are some of the most advanced in the world. Israel's missile and air defense systems include four independent components, each serving different purposes for air and missile defense. The Arrow air defense system, which includes the Arrow, the Arrow 2, and Arrow 3, is Israel's long-range air defense system. The air defense system operates outside of the atmosphere, and is designed to intercept long-range ballistic missiles. Israel's David's Sling air defense system is meant to intercept medium-range missiles. Israel's Iron Dome is developed and designed to protect Israel against rockets and short-range missiles. Finally, the C-DOME is the maritime version of the Iron Dome. The C-DOME is an agile, quick-reaction, small-footprint ship-based air defense system that protects combat vessels against a wide range of maritime and coastal threats.

4 Scenario Analysis

In the event that Tehran achieves a nuclear weapons capability, the United States must have well thought out plans for how to respond to a variety of Iranian threats and circumstances in the Middle East. However, it is critical to understand that the deterrence and deescalation calculus of the United States is, and must be, different given varying scenarios. The United States must understand

³⁹Missile Defense Project, "Missiles of Israel," Missile Threat, Center for Strategic and International Studies, June 14, 2018.

the possible scenarios across the spectrum of crisis, conflict, and escalation, and also be well prepared to respond and conduct its diplomatic and military operations in accordance with each scenario. What are the possible scenarios the U.S. might be forced to confront in the Middle East given a nuclear weapons capable Iran?

It is important to note that the deterrence and deescalation calculus of the U.S. is necessarily different from that under the current circumstances. While containment of Iran will remain at the core of U.S. policy as long as Iran continues to subvert and threaten its neighbors either directly or through its proxies whether or not Iran possesses a nuclear arsenal, the U.S. might face far greater challenges in doing so if Iran achieves a weapons capability. A combination of containment policies, effective deterrence, sanctions and expressions of soft power will likely be necessary to confront a nuclear Iran. While one of the greatest threats a nuclear Iran poses would be the use of its capabilities against Israeli and U.S. assets in the Middle East, perhaps the largest threat to stability in the Middle East is that Iran's nuclear capability would serve as an effective deterrent against Israel and the U.S. Since Iran's nuclear arsenal would likely remain far inferior to that of its adversaries, it seems unlikely that Iran would use its nuclear capability in any capacity short of regime survival. However, its nuclear capability might allow Tehran to pursue an even more belligerent and outwardly aggressive strategy against its adversaries than we currently see. Iran's nuclear deterrent would depend largely on its ability to inflict massive damage against its adversaries in response to an attack against its most important infrastructure or population centers. Thus, it seems that the most effective defense against a nuclear Iran would involve a strategy of "deterrence by denial." Deterrence by denial relies on physically denying Iran the capacity to conduct a successful attack against its adversaries, or at least drastically diminishing the capacity of Iran to inflict damage upon its adversaries. However, this policy might also necessitate a supplementary and distinct kind of deterrence— "deterrence by punishment." This kind of deterrence is dependent on inflicting severe damage upon the adversary in response to an attack— that is to say, retaliation.

4.1 Peacetime

The prospect of any extended, or even brief period of peace in the Middle East between Israel and Iran and its proxies seems more unlikely by the day. Certainly, the prospects for such a scenario seems even less likely in the event of a nuclear armed Iran. However, if there is a period of relative peace, it would be an essential time for the U.S. and Israel to prepare for what would likely be an inevitable coming crisis or conflict. A peaceful period of time would also be one in which the U.S. could act to establish deterrence against both Iran and Israel from aggressive military action against the other. To be clear, "peacetime" would involve an extended period in which Iranian proxies do not launch attacks against Israel, Iran does not launch attacks on Israel, and there are no conflicts between Israel and Iranian proxies or Iran itself. Thus, a lack of any offensive measures

by Israel against Iran or its proxies, including aerial attacks with missiles or drones on Iranian or proxy targets, even those targeting individual political or military leaders of Iran or its proxies, would be necessary components of peacetime.

4.2 Crisis

Crises scenarios in the Middle East pose major risks for escalation to conventional war, flurries of missile and other aerial attack exchanges, and the potential for nuclear use. Oran R. Young generally agrees with Charles McClelland's definition of international crises: "An international crisis is a set of rapidly unfolding events which raises the impact of destabilizing forces in the general system or any of its sub-systems substantially above 'normal' (i.e., average) levels and increases the likelihood of violence occurring in the system."⁴⁰ Due to the adversarial dynamic between Israel and Iran, communications between Tehran and Jerusalem are all but absent, especially in times of crisis between the two states. Uncertainty about the intentions, actions, and plans of the adversary cause crisis scenarios to be incredibly unstable, with potential for rapid and violent escalation. Thus, the U.S. must be prepared to involve itself quickly and effectively in a range of crisis scenarios in order to prevent unacceptable escalation. Rapid and effective U.S. involvement in crisis scenarios is currently essential, and in scenarios involving a nuclear Iran, the necessity for such U.S. involvement would be dramatically greater.

While Iran and Israel are reticent to engage in direct conflict with each other, Iran is willing to engage in and support aggressive attacks against Israel through its proxies. Notably, while Iran's proxies align ideologically with Tehran, and respond to the orders of Tehran to some, but varying degrees, Iran does not exercise complete control over the actions of its proxies. Thus, as Alterman et. al point out, "working with partially autonomous actors can pose a liability at times for Iranian leaders, especially in times of crisis when rapid decisions are needed"⁴¹ The actions of proxies both supported and unauthorized by Tehran against Israel create perilous circumstances with significant risks of further escalation. Crisis scenarios include those that have arisen constantly throughout the current conflict between Israel and Hamas, including violent exchanges between Israel and other Iranian proxies. If Israel were to launch a large attack on Iranian proxies, such as a devastating attack on Hezbollah targets in Lebanon in Syria, and perhaps a combined air assault and ground invasion, a severe crisis scenario between Iran and Israel would arise. These examples raise the risks of conflict between Iran and Israel to high levels, creating crisis scenarios between Iran and Israel. Persistent attacks by other Iranian proxies, such as Hezbollah missile, rocket, and drone attacks on Northern Israel also create crisis scenarios between Iran and Israel. Other scenarios involve those in which Israel conducts targeted decapitation operations on high ranking Iranian,

⁴⁰Oran R. Young, *The Intermediaries: Third Parties in International Crises* Princeton University Press, 1967

⁴¹Alterman, Jon B., Michael Connell, Michael Eisenstadt, Farideh Farhi, Thomas Karako, J. Matthew McInnis, Hijab Shah, et al. 2017. "Deterring Iran after the Nuclear Deal," pg. 42.

and Iranian proxy military and political leaders. These Israeli operations are common occurrences. Examples of such operations include those in the latter weeks of July 2024, where Israel has killed several high-profile Hamas and Hezbollah leaders in quick succession, prominently Hamas leader Ismail Haniyeh in Tehran and Fuad Shukr, whom Israel called Hezbollah's most "senior military commander." Other crisis scenarios arise from exchanges of limited strikes and missile assaults, and retaliatory strikes directly between Iran and Israel. For example, a series of strikes and counter-strikes between Israel and Iran in April 2024, including the first ever direct attack on Israel from Iranian soil, heightened tensions between Iran and Israel. On April 13, Iran initiated the first-ever direct military action against Israel from Iranian territory. Iran launched a series of aerial attacks including drones and ballistic and cruise missiles against Israel. These attacks were publicly stated to be retaliation for an April 1 strike reportedly by Israel against a building within the Iranian embassy compound in Damascus, Syria, which killed leaders of the IRGC. In response to the attack by Iran, on April 19, Israel launched an attack near an Iranian military base in the central province of Isfahan. Although the attack appeared to be relatively narrow in scope, it may have signaled an Israeli ability to evade and target Iranian air defenses—apparently damaging the radar on a Russian origin S-300 system—in a province where some of Iran's nuclear facilities are located.⁴² This limited exchange caused severe risks of further escalation, and although further escalation was avoided, such scenarios arising from exchanges like the one above certainly constitute crisis scenarios. Crisis scenarios might also arise without any military attacks. For example, either Iran or Israel might receive intelligence that the other is readying its nuclear weapons delivery systems, deploying its nuclear weapons capabilities in forward positions, or conducting nuclear or ballistic missiles launch tests. Due to the lack of communication between Tehran and Jerusalem, the large degree of uncertainty concerning the other state's intentions and plans in such a scenario might lead to rapid escalation through preemptive measures.

One of the distinct dangers of many of the crisis scenarios described above is that each may quickly escalate to the level of a direct conventional conflict between Iran and Israel. This may seem intuitive, as the risks of crisis scenarios involve escalation to conventional conflict, but the possible mechanisms for such escalation are worth noting. There are a variety of possible mechanisms for escalation from crisis to conflict scenarios: The first possible mechanism (and arguably most likely and dangerous) involves Iranian proxies launching damaging attacks against Israel's air defenses. This could lead to Israel becoming far more vulnerable to attacks from proxies and Iran. In such a case, Israel might deem sustaining proxy attacks untenable, but may or may not find a multi-front ground war against proxies to be nonviable. Then, Israel might conduct direct attacks against Iran to deter further Iranian support for proxy attacks against Israel, and, anticipating retaliation, Israel might strike with a sweeping counter-force attack (against Iranian nuclear sites, missile

⁴²Zanotti, Jim. "Israel: Major Issues and U.S. Relations."

launch sites, air force bases, air defense systems, and military infrastructure). Such a sweeping attack is almost certainly not 100% effective, and would very likely lead to Iran retaliating with major military initiatives, and possibly escalation to the nuclear level. In the case that Israel or Iran receives intelligence of the other deploying its nuclear weapons capabilities in forward positions, or conducting nuclear or ballistic missiles launch tests, there is another mechanism for escalation. Either Israel or Iran – the defender in the scenario – might deem the actions of the adversary to be a *casus belli*, and in turn might launch major preemptive counter-force attacks against the other. If Israel conducts repeated attacks on Iran proxy leaders, or Iranian officials in Iran, Iran might retaliate with major missile attacks against Israel, as it has done recently. While recent escalation has managed to subside in these cases, it is no guarantee that the same will happen in the future, especially if Iran possesses a nuclear weapons capability. Thus, Israel might respond in kind to Iranian attacks, and rapid escalation could follow. If Israel or Iran launches missile strikes that damage the other's nuclear sites, air defenses, or missile launch sites, escalation to unrestrained conflict could follow. Since uncertainty is prevalent without communication between the two sides, the defending side might perceive the attack as threatening subsequent large scale operations, due to their own diminished ability to defend itself, or retaliate against future attacks. Then, the vulnerable side could decide to launch its own attack (facing a "use it or lose it" dilemma), before its defensive and offensive capabilities have been incapacitated to unacceptable levels. Finally, if Iran continues to follow its recent precedent of launching direct attacks against Israel's "Zone of the Interior" (previously unprecedented), Israel might also deem the kind of attack to be a *casus belli*, or at least an unsustainable precedent to follow and endure. Thus Israel could conclude that direct conflict is the only option to decisively achieve victory, rather than continuing to endure limited costly Iranian attacks. In this case, Israel might engage all of its military capabilities in fighting, as its military doctrine suggests, in order to consolidate and deploy a rapid and conclusive war effort against Iran, including nuclear use for counter-force or even counter-value purposes. Any of these scenarios in which crisis escalates to all out conflict would likely result in an incredibly long, violent, and costly hot war between Iran and Israel, although the U.S. might be able to intervene to deescalate at various stages of conflict, which vary given the different kinds and stages of conventional conflict.

4.3 Limited/Conventional Conflict

Limited or Conventional conflicts between Israel and Iran raise tensions to dangerous levels, which may lead to nuclear escalation. De-escalatory involvement and policy by the U.S. would be essential to prevent a conventional conflict from spiraling into an all out nuclear exchange in the Middle East. While these scenarios are relatively unlikely, due to the aforementioned aversion of both Israel and Iran to engage in direct conflict with the other, the risks associated with such a conventional

conflict are immense. Unlike crisis scenarios, which are isolated and separate incidents (i.e. an attack and retaliation then subsiding tensions), a conflict scenario must involve persistent exchanges and attacks, with less publicly declared motivation and communication of intentions. Many of the examples of attacks and military operations which lead to crisis scenarios have historically been accompanied by public statements of intent and justification. However, limited conventional conflict is unlikely to be accompanied by the same forewarning of attacks and operations. Other criteria for a conventional conflict would include repeated and ongoing aerial attacks and missile exchanges between Iran and Israel. Of course, conventional conflict includes scenarios in which ground forces are deployed for invasion and combat.

4.4 Nuclear Use

One of the most dangerous scenarios imaginable in the Middle East is one in which the nuclear taboo is breached— either by Israel or Iran. It seems far more likely that nuclear use would be initiated by Israel. There are a few different kinds of nuclear strikes that Israel might possibly employ. The first, and most likely, could be the firing of a non-lethal nuclear demonstration shot over a remote location in Iran. Less likely, albeit still possible, would be the launch of a “precision” nuclear strike of weapons against Iranian military targets, aiming to cripple Iranian offensive forces and perhaps induce enough chaos to prompt the Iranian revolutionary regime to collapse. The most severe possibility would be an Israeli nuclear attack against Tehran, or broadly counter-force and counter-value targets, maximizing damage, casualties, and strain on the Iranian regime. Although these potential scenarios are ordered in terms of likelihood, it is important to note that the scenarios also fall along different stages of the escalation ladder. It is unlikely (albeit not impossible) that Israel would initiate nuclear use against a range of counter-force, or even counter-value targets before deploying a demonstrative strike. Similarly, while it is far more likely that Israel would be the party in any scenario to initiate nuclear use, it is likely (although not certain) that Iran would respond to any breach of the nuclear taboo by Israel with its own nuclear use. That is to say, if Israel were to deploy a demonstrative nuclear strike, Iran might respond with its own nuclear strike in the southern Negev Desert in Israel, or against important counter-force targets in Israel. Additionally, if Israel were to launch a large nuclear counter-force attack on critical Iranian military infrastructure, air defense, and missile launch sites, Iran might respond in kind. In any of these scenarios, it is essential that the United States have contingency plans to deescalate, and prevent spiraling nuclear escalation in the Middle East at any stage of the nuclear escalation ladder.

5 U.S. Policy Recommendations

The United States maintains several key defensive commitments and policies to ensure the security and stability of Israel. These commitments are characterized by military aid, joint defense initiatives, diplomatic support, and strategic cooperation. The U.S. provides Israel with substantial military aid through Foreign Military Financing (FMF). Under the 2016 Memorandum of Understanding (MOU), the U.S. committed \$38 billion in military aid to Israel over ten years (2019-2028), which includes \$33 billion in FMF grants and \$5 billion for missile defense programs.⁴³ Annual aid is primarily used for procuring advanced military equipment, maintaining Israel's qualitative military edge (QME), and ensuring its self-defense capabilities. The U.S. has provided significant funding and technical support for Israel's Iron Dome missile defense system, which intercepts short-range rockets and artillery shells. The U.S. collaborates with Israel on other missile defense systems, including David's Sling (for mid-range threats) and the Arrow series (for long-range ballistic missiles). The two countries engage in joint military exercises and the co-development of defense technologies to enhance missile defense and broader security cooperation. The U.S. and Israeli military's regularly conduct joint training exercises to improve interoperability and readiness. These exercises cover a wide range of scenarios, including air defense, cyber defense, and counter-terrorism operations. The U.S. maintains stockpiles of military equipment in Israel for use by U.S. forces in the region and, in emergencies, by Israeli forces. The U.S. and Israel have a close intelligence-sharing relationship, focusing on mutual threats such as terrorism, nuclear proliferation, and regional adversaries. The U.S. consistently supports Israel in international forums, including the United Nations, often vetoing resolutions perceived as unfairly critical of Israel. U.S. policy is committed to maintaining Israel's QME, meaning Israel must have superior military capability relative to potential regional adversaries. This policy ensures Israel's ability to defend itself against any coalition of adversarial states. The U.S. is prepared to provide additional military support to Israel in times of crisis. This includes emergency munitions supplies and other forms of rapid assistance. The U.S. and Israel have developed contingency plans for various scenarios, ensuring coordinated responses to threats. Regular high-level meetings between U.S. and Israeli officials address strategic concerns, regional developments, and bilateral security cooperation. Various defense agreements formalize the commitments and cooperation between the two countries, ensuring a structured approach to mutual security. These commitments reflect the deep strategic partnership between the United States and Israel, aimed at ensuring Israel's security, maintaining regional stability, and addressing shared threats. However, the U.S. has no mutual defense treaty with Israel, and its commitments to Israel's defense in the event of a conflict remain ambiguous.

The United States does not have a formal mutual defense pact with Israel for several reasons, despite their close and strategic partnership. There are a few reasons for a lack of such a defense

⁴³"U.S. Relations with Israel." United States Department of State.

pact. A formal mutual defense treaty could potentially limit the flexibility of the United States and Israel to respond independently to threats and regional dynamics. Both countries prefer to maintain the ability to act unilaterally or through informal cooperation as needed. A formal defense treaty with Israel could exacerbate tensions in the Middle East. Many Arab and Muslim-majority countries already view the U.S.-Israel relationship with suspicion, and a formal defense pact could further complicate U.S. relations with these countries and hinder peace efforts. The U.S. and Israel have extensive security cooperation without a formal treaty. This includes significant U.S. military aid, joint training exercises, intelligence sharing, and defense technology collaboration. The U.S. provides Israel with advanced weaponry and missile defense systems, such as the Iron Dome. Various U.S. laws and policies ensure Israel's security, such as the 2016 U.S.-Israel Memorandum of Understanding (MOU). Israel prefers to maintain its defense autonomy. A mutual defense pact could require Israel to seek U.S. approval for military actions, potentially limiting its ability to respond quickly and decisively to threats. The U.S. has numerous global military commitments, and a formal defense pact with Israel could further stretch U.S. military resources and obligate the U.S. to engage in conflicts that might otherwise be avoided or managed differently. While there is strong bipartisan support for Israel in the U.S. Congress, a formal mutual defense treaty might face political and public scrutiny. Some lawmakers and constituents might oppose further entangling the U.S. in Middle Eastern conflicts. While there is no formal treaty, U.S. leaders have consistently affirmed their commitment to Israel's security. High-level diplomatic and military dialogues ensure that both countries are aligned on defense and security matters. Overall, the U.S. and Israel maintain a robust and multifaceted security partnership that effectively addresses mutual defense needs without the formal structure of a mutual defense pact. This arrangement provides both nations with the flexibility and autonomy to navigate the complex geopolitical landscape of the Middle East.

The United States has many strategies at its disposal in confronting a nuclear Iran. Certain strategies would likely offer the best chances for the United States to deter a nuclear Iran from unacceptable aggression in the Middle East, assure Israel of U.S. support, and allow the United States to rapidly and effectively respond to threats, crises, and conventional conflict scenarios in the Middle East with de-escalatory action.

1. *Permanent naval presence in the Eastern Mediterranean*

The United States should establish a constant naval presence in the Eastern Mediterranean. This strategy would involve the permanent stationing of naval ships in the region, enhancing the U.S. ability to project power and respond quickly to any threats. A continuous naval presence would deter Iranian aggression and provide reassurance to U.S. allies in the region. Throughout the past few months, the United States has rotated stationing ships in the Mediterranean, off the coast of Israel, primarily in response to heightened threats of attacks

on Israel from Iran. Many of these decisions have been reactive, and have been effected in periods after Israeli offensive operations against Iran and its proxies, or when Iran publicly threatens and warns Israel of impending aerial attacks. By maintaining a constant rotation of ships in the Mediterranean, the United States would be better situated to quickly respond to the threat of attack by a nuclear Iran. This would differ from current policy, which is largely decided on a case by case, or situational basis. A continuous naval presence would show a commitment to the defense of Israel from an Iranian attack.

To reinforce defensive commitments to the defense of Israel, the U.S. should include provisions for permanent military stationing initiatives. These provisions would accompany and support continuous naval stationing initiatives. This would entail the establishment of enduring bases or rotational deployments of U.S. forces in allied countries. Such measures would demonstrate a steadfast commitment to the defense of these nations and enhance collective security arrangements.

2. Comprehensive denial and punishment campaign against Iranian proxies

The United States should sustain and intensify a comprehensive denial and punishment campaign aimed at neutralizing Iranian proxies. This involves a combination of military, economic, and diplomatic actions designed to disrupt the operations and influence of groups supported by Iran. Continuous intelligence gathering, targeted sanctions, and direct military action where necessary are key components of this strategy. The United States currently has taken extensive measures to hinder Iranian support for its proxies, and has struck back at Iran's proxies in response to persistent attacks against U.S. assets in the Middle East. However, the current military strategy employed by the United States, which is largely reactive, might prove insufficient to combating the potentially heightened aggression of Iran's proxies if Tehran achieves a nuclear weapons capability. Thus, a nuclear Iran might necessitate a more proactive version of the United State's existing strategy against Iranian influence and aggression through proxies. A more proactive version of the United State's current strategy would involve a combination of punishment through sanctions, destroying supply chains, limiting arms transfers, and striking critical military targets, capabilities, and personnel of Iranian proxy forces. By employing these tactics continuously and proactively, the United States and its partners may be able to diminish the threat posed to U.S. and Israeli interests from Iranian proxies, and in turn reduce Tehran's influence, military capabilities, and capacity for outward aggression in the Middle East.

3. Bolstering Israeli air defense

The U.S. should continue to bolster Israeli air defenses through robust military aid packages and arms sales. This support must address specific vulnerabilities in Israel's defense sys-

tems against attacks from Iranian proxies and direct assaults from Iran. Enhancements could include advanced missile defense systems, early warning technologies, and increased joint military exercises to ensure preparedness and interoperability. Israel faces a significant threat from Iran's growing arsenal of ballistic missiles, which have been steadily enhanced in range, precision, and payload capacity. Moreover, Iran's proxies, particularly Hezbollah, possess an extensive inventory of rockets, missiles, including precision-guided munitions capable of overwhelming Israel's missile defense systems through saturation attacks. The U.S. should prioritize the enhancement of Israel's existing missile defense architecture. This includes the Iron Dome for short-range threats, David's Sling for medium-range missiles, and the Arrow system for long-range ballistic missile threats. Additional funding and technological upgrades should focus on increasing interception success rates, especially under scenarios involving large-scale, simultaneous barrages. Enhancing the resilience and redundancy of these systems against cyber threats and electronic warfare is also essential. Iranian ballistic missile capabilities, coupled with the potential for surprise attacks by Iranian proxies, underscore the need for advanced early warning systems. The detection of missile launches and the ability to track incoming threats in real-time are critical for effective interception and civilian protection. The U.S. should assist Israel in deploying next-generation early warning radars and satellite-based surveillance systems. These enhancements will reduce response times and improve the accuracy of threat assessments. Integration with U.S. intelligence assets, including satellite imagery and signals intelligence, will further refine Israel's ability to anticipate and neutralize threats before they materialize. Iran and its proxies have increasingly utilized UAVs for reconnaissance, targeted strikes, and swarm attacks. These UAVs pose a unique challenge due to their ability to evade traditional radar systems and their potential to be deployed in large numbers. The U.S. should provide Israel with cutting-edge counter-UAV systems, including directed energy weapons and advanced electronic warfare capabilities to detect, jam, and neutralize UAV threats. Enhancing Israel's network of integrated air defense systems to specifically counter the UAV threat will be critical in mitigating this growing vulnerability. By specifically targeting these vulnerabilities, the U.S. can significantly enhance Israel's defensive posture, ensuring that it remains capable of withstanding and neutralizing the multi-faceted threats posed by Iran and its proxies. This strategic bolstering of Israel's defenses will also serve to deter aggression by demonstrating that Israel's defense systems are both robust and technologically advanced.

4. *Explicit defensive commitments to Israel*

The United States should make explicit defensive commitments to Israel. This would involve a public and unequivocal declaration of support, emphasizing that U.S. assistance is strictly for defensive purposes. Such commitments could include advanced defensive capabilities,

intelligence sharing, and coordinated response plans to counter any potential threats. These commitments most notably should declare unwavering dedication of United States military systems to engage and prevent missile and other aerial attacks from Iran against Israel. By making these commitments explicit, the U.S. would send a strong message of deterrence to Iran and reinforce the security of its ally, Israel.

While the U.S. consistently reaffirms its commitment to Israel's security, much of this support is couched in broad terms that lack specificity, particularly concerning the U.S. military's role in defending Israel against direct attacks. Statements of support often emphasize the U.S. commitment to maintaining Israel's qualitative military edge (QME) but stop short of clearly articulating the extent of U.S. involvement in an actual conflict scenario. The absence of explicit guarantees creates strategic ambiguity, which, while sometimes useful, can also lead to miscalculations by adversaries like Iran. Without clear and public declarations of U.S. military commitments, there is a risk that Iran might perceive an opportunity to test Israel's defenses without provoking a full U.S. military response.

The U.S. should make a clear and public declaration that it is committed to defending Israel against any direct attacks, particularly from Iran. This declaration should specify that U.S. military assets, including missile defense systems, air defense capabilities, and intelligence resources, will be actively employed to intercept and neutralize any threats before they reach Israeli territory. Such a declaration would remove any ambiguity about the U.S. role in Israel's defense and serve as a powerful deterrent against potential Iranian aggression. The U.S. should make an explicit commitment to deploying its advanced missile defense systems, such as Aegis-equipped ships, THAAD, and Patriot batteries, across the Middle East with the express purpose of intercepting and destroying any missile or aerial threats launched by Iran towards Israel. This commitment would involve utilizing all relevant U.S. military infrastructure in the region, including bases in the Gulf and Mediterranean, to ensure a layered and comprehensive missile defense shield that extends beyond Israel's borders. The U.S. should enhance its provision of advanced defensive technologies to Israel, including systems designed to counter emerging threats such as hypersonic missiles and advanced UAVs. This would also involve greater integration of U.S. and Israeli missile defense networks, ensuring seamless communication and coordination during a crisis. Additionally, the U.S. should commit to real-time intelligence sharing with Israel, particularly concerning Iranian missile launches, troop movements, and potential cyber threats. This would enable Israel to respond more quickly and effectively to any imminent threats. The U.S. and Israel should also develop and publicize coordinated military response plans for various scenarios, including large-scale missile attacks from Iran, proxy attacks by Hezbollah, and combined conventional and cyber assaults. These plans should outline the specific roles of U.S. and

Israeli forces, ensuring a unified and efficient response that maximizes the defensive capabilities of both nations. Regular joint military exercises that simulate these scenarios would not only enhance readiness but also demonstrate to Iran and other adversaries the depth of U.S.-Israeli military cooperation. By making these commitments explicit, the United States would not only reinforce the security of its ally, Israel, but also send a clear message of deterrence to Iran. These measures would signal that any attempt to attack Israel would be met with an immediate and overwhelming defensive response, utilizing the full spectrum of U.S. military capabilities in the region. This would reduce the likelihood of miscalculation and contribute to greater stability in the Middle East, which would be essential in confronting a nuclear Iran.

6 Conclusion

The strategies currently in place, along with the proposed enhancements, aim to deter Iran from achieving nuclear weapons capability. However, despite Iran's participation in the Nuclear Non-Proliferation Treaty (NPT), the International Atomic Energy Agency (IAEA) has limited enforcement capabilities, which constrains its ability to prevent Iran from advancing its nuclear program.

Iran has already achieved nuclear latency, meaning it possesses the knowledge and infrastructure necessary to produce nuclear weapons. This capability allows Iran to potentially make the transition to a nuclear-armed state without the United States and its partners being fully aware. The threat of a nuclear-armed Iran is significant, and despite the best efforts of the international community, it may not be possible to fully stop Iran from crossing this threshold. The U.S. must therefore be prepared for the eventuality of a nuclear Iran.

As the geopolitical landscape shifts, so too must the U.S. approach to deterrence and de-escalation. The calculus for responding to a nuclear Iran varies depending on the circumstances, requiring tailored strategies for different scenarios. There are specific policies the U.S. can implement now and in the future to better prepare for this looming threat. These measures will equip defense and military officials with the tools needed to develop comprehensive contingency plans for any crisis, conflict, or nuclear scenario in the Middle East. By doing so, the U.S. can strengthen its position and ensure it is ready to protect its interests and allies in a region increasingly shaped by the potential for a nuclear-armed Iran.